Custom 16-bit RISC SoC for Digital Lock with VGA Status Display

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***Abstract*— This project proposes the design and implementation of a custom 16-bit RISC system on chip on the Nexys A7 100-T Board. The system will include a simple processor core, instruction ROM, data RAM and memory-mapped peripherals for switches, push buttons, LEDs, seven segment displays, and VGA output. The processor will execute a digital lock application in which a user enters a code through board inputs, and the system responds by displaying lock status messages such as locked, access granted, access denied, and lockout. The purpose of this project is to demonstrate processor design, memory interfacing, peripheral control, and hardware/software interaction in a visible and interactive embedded environment. The completed design will be verified through simulation of individual modules and demonstrated on the Nexys A7 board with real time visual output.**

I. Introduction

This project implements a custom 16-bit RISC processor on the Nexys A7-100T FPGA board to control a digital lock system with VGA status output. The system demonstrates processor design, memory-mapped input/output, embedded control, and hardware verification. The processor executes a program stored in ROM to read user inputs, compare passcodes, update lock state, and drive peripheral outputs including LEDs, seven segment display, and VGA.

II. Background & Preliminaries

This project is centered on the design of a custom 16-bit RISC System-on-chip (SoC) implemented on the Nexys A7-100T FPGA board for a digital lock application with VGA status output. The overall system combines a simple processor core, instruction ROM, data RAM, and memory-mapped peripherals so that the lock behavior can be controlled in software while still being demonstrated through real hardware. In this design, the user enters a code through the board inputs, and the system responds by updating LEDs, seven-segment displays, and a VGA monitor with status messages such as LOCKED, ACCESED GRANTED, ACCESS DENIED and LOCKOUT. This makes the project a practical example embedded digital system design, since it integrates processor architecture, peripheral interfacing, and visble user feedback into one FPGA-based platform

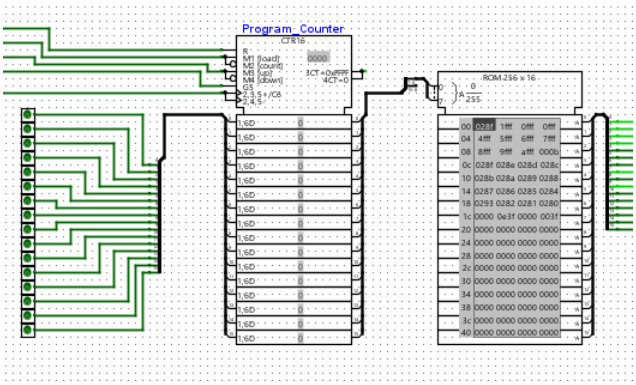
III. Implementation

A. Top-Level Implementation of 16-bit RISC SoC

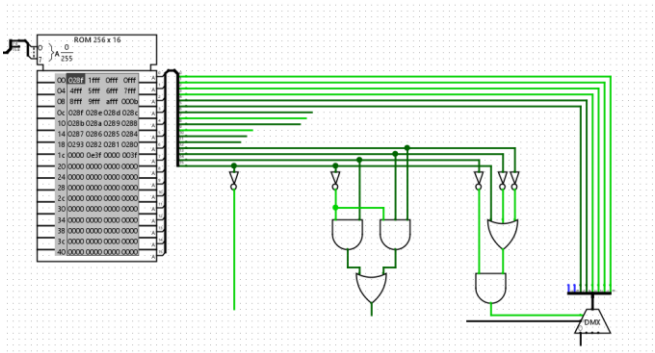
The Current implementation of the project was developed as an early functional prototype of the cutom 16-bit RISC SoC. At this stage, the design is organized as a single top-level circuit that combines the processor datapath, instruction memory, data memory, control selection logic, and memory-mapped peripheral registers. This prototype is being used to verify the system architecture before full FPGA/Vivado integration. This aligns with the project goal of building a simple processor core with ROM, RAM, and memory-mapped peripherals for user interaction and output control.

B. Program Counter + Instruction ROM

The processor portion of the circuit includes a 16-bit program counter, an instruction ROM, and a collection of general-purpose 16-bit registers labeled through R0 through R7.



The arithmetic and logic functionality is implemented using dedicated 16-bit adder, subtractor, and compartor, AND, OR, and NOT logic blocks. These components provide the core operations needed for the lock-control application, such as comparing entered values, updating state, and generating control-dependent results. Together, these blocks serve as the working ALU portion of the processor prototype



C. Register file area

For memory support, the circuit contains both instruction ROM and data RAM. The instruction ROM stores the current machine-level program as hexadecimal values, while

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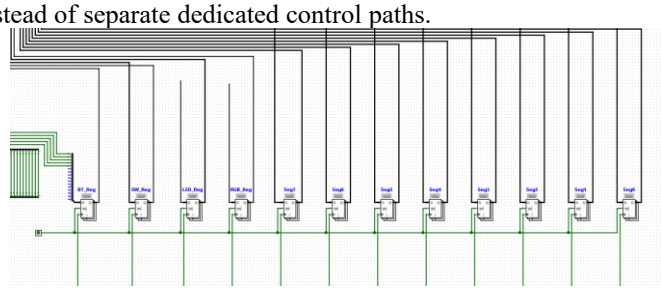
the RAM is reserved for temporary values and internal state storage. In addition, the design includes several memory-mapped peripheral registers, including BT-Reg for button input, SW_Reg for switch input, LED_Reg for LED output, RGB_Reg for RGB LED output, and Seg7 through Seg0 for seven-segment display output. This follows the intended memory-mapped I/O approach, where the CPU communicates with peripherals through register addresses instead of separate dedicated control paths.



D. VGA (Coming soon)

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	Table column subhead	***Subhead***	***Subhead***
copy	More table copy[a]		

[a] Sample of a Table footnote. (*Table footnote*)

Fig. 1. Example of a figure caption. (*figure caption*)

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[1] G. Eason, B. Noble, and I. N. Sneddon, "On certain integrals of Lipschitz-Hankel type involving products of Bessel functions," Phil. Trans. Roy. Soc. London, vol. A247, pp. 529–551, April 1955. *(references)*

[2] J. Clerk Maxwell, A Treatise on Electricity and Magnetism, 3rd ed., vol. 2. Oxford: Clarendon, 1892, pp.68–73.

[3] I. S. Jacobs and C. P. Bean, "Fine particles, thin films and exchange anisotropy," in Magnetism, vol. III, G. T. Rado and H. Suhl, Eds. New York: Academic, 1963, pp. 271–350.

[4] K. Elissa, "Title of paper if known," unpublished.

[5] R. Nicole, "Title of paper with only first word capitalized," J. Name Stand. Abbrev., in press.

[6] Y. Yorozu, M. Hirano, K. Oka, and Y. Tagawa, "Electron spectroscopy studies on magneto-optical media and plastic substrate interface," IEEE Transl. J. Magn. Japan, vol. 2, pp. 740–741, August 1987 [Digests 9th Annual Conf. Magnetics Japan, p. 301, 1982].

[7] M. Young, The Technical Writer's Handbook. Mill Valley, CA: University Science, 1989.

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